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1 Partial Derivatives

We first consider taking the derivative of a multivariable function "one variable at a time". More precisely, if we have a function $f(x,y)$ of two variables, then by holding y fixed at a value $y=b$, then $F(x)=f(x,b)$ is a function of one variable. We can then take the derivative $\frac{dF}{dx}$ with respect to x. Similarly, we can hold $x=a$ fixed to obtain a function $G(y)=f(a,y)$ and take the derivative $\frac{dG}{dy}$ with respect to y.

Definition 1.1 (Partial derivative)**.** Let $f: U \subset \mathbb{R}^2 \to \mathbb{R}$ be defined on an open subset of $\mathbb{R}^2$ containing the point (x,y). Then the real number

$$\frac{\partial f}{\partial x}(x,y) = \lim_{h\to 0} \frac{f(x+h,y)-f(x,y)}{h}$$

is called the *partial derivative of f with respect to x at the point* (x,y), provided this limit exists. Similarly, the real number

$$\frac{\partial f}{\partial y}(x,y) = \lim_{h\to 0} \frac{f(x,y+h)-f(x,y)}{h}$$

is called the *partial derivative of f with respect to y at the point* (x_0,y_0), provided this limit exists.

More generally, if $f: U \subseteq \mathbb{R}^n \to \mathbb{R}$, then $\frac{\partial f}{\partial x^i}$ is defined by

$$\frac{\partial f}{\partial x^i} = \lim_{h\to 0} \frac{f(x_1,\ldots,x_i+h,\ldots,x_n)-f(x_1,\ldots,x_i,\ldots,x_n)}{h}.$$

Note that each $\frac{\partial f}{\partial x^i}$ is a real-valued functions on U (or some proper subset of U if it is not defined at all points of U). Another notation for the partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ are f_x and f_y, respectively.

Example 1.2. If $f(x,y)=x^3+x^2y^3-2y^2$, then

$$\frac{\partial f}{\partial x} = 3x^2+2xy^3$$
$$\frac{\partial f}{\partial y} = 3x^2y^2-4y.$$

Exercise 1.3. Calculate the partial derivatives with respect to x and y of the function $f(x,y)=x^2y+\cos(x+y)$.

$$\frac{\partial f}{\partial x} = 2xy-\sin(x+y)$$
$$\frac{\partial f}{\partial y} = x^2-\sin(x+y).$$

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Exercise 1.4. Calculate the partial derivatives with respect to x and y of the function $g(x,y) = \dfrac{xy}{x^2+y^2}$.

$$\frac{\partial g}{\partial x} = \frac{y(y^2-x^2)}{(x^2+y^2)^2},$$
$$\frac{\partial g}{\partial y} = \frac{x(x^2-y^2)}{(x^2+y^2)^2}.$$

By definition, the partial derivative is the instantaneous rate of change of f when all variables, except for the specified one, are held fixed. In the case where f is a function of two variables, we can understand $\frac{\partial f}{\partial x}(a,b)$ geometrically as the slope of the tangent line to the curve obtained by intersecting the surface $z = f(x,y)$ with the plane $y = b$ at the point $(a,b,f(a,b))$.

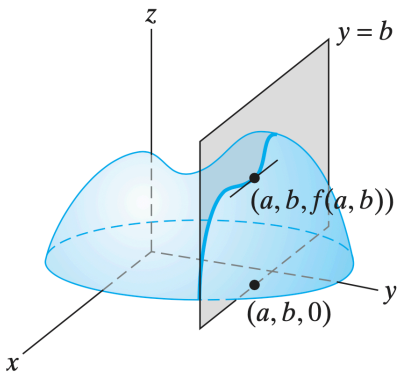


Figure 1: Visualizing the partial derivative $\frac{\partial f}{\partial x}(a,b)$.

Similarly, $\frac{\partial f}{\partial y}(a,b)$ is the slope of the tangent line to the curve obtained by intersecting the surface $z = f(x,y)$ with the plane $x = a$ at the point $(a,b,f(a,b))$.

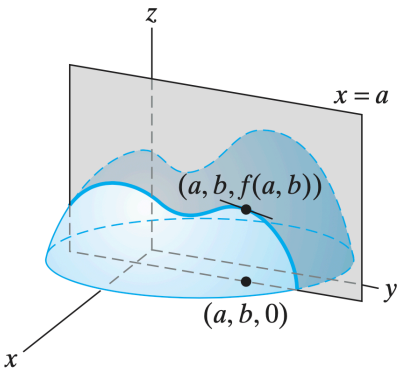


Figure 2: Visualizing the partial derivative $\frac{\partial f}{\partial x}(a,b)$.

EXAMPLE 3 Occasionally, it is necessary to appeal explicitly to limits to evaluate partial derivatives. Suppose $f: \mathbf{R}^2 \to \mathbf{R}$ is defined by

$$f(x, y) = \begin{cases} \dfrac{3x^2y - y^3}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}.$$

Then, for $(x, y) \neq (0, 0)$, we have

$$\frac{\partial f}{\partial x} = \frac{8xy^3}{(x^2 + y^2)^2} \quad \text{and} \quad \frac{\partial f}{\partial y} = \frac{3x^4 - 6x^2y^2 - y^4}{(x^2 + y^2)^2}.$$

But what should $\dfrac{\partial f}{\partial x}(0, 0)$ and $\dfrac{\partial f}{\partial y}(0, 0)$ be? To find out, we return to Definition 3.2 of the partial derivatives:

$$\frac{\partial f}{\partial x}(0, 0) = \lim_{h \to 0} \frac{f(0 + h, 0) - f(0, 0)}{h} = \lim_{h \to 0} \frac{0 - 0}{h} = 0,$$

and

$$\frac{\partial f}{\partial y}(0, 0) = \lim_{h \to 0} \frac{f(0, 0 + h) - f(0, 0)}{h} = \lim_{h \to 0} \frac{-h - 0}{h} = \lim_{h \to 0} -1 = -1. \quad \blacklozenge$$

Exercise 1.5. If $f(x,y) = \sin(\frac{x}{1+y})$, calculate $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial x}$.

$$\frac{\partial f}{\partial x} = \cos(\frac{x}{1+y})\frac{1}{1+y}, \quad \frac{\partial f}{\partial x} = -\cos(\frac{x}{1+y})\frac{x}{(1+y)^2}$$

Exercise 1.6. If $f(x,y) = y\sin(xy)$, calculate $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial x}$.

$$\frac{\partial f}{\partial x} = y^2\cos(xy), \quad \frac{\partial f}{\partial y} = \sin(xy) + xy\cos(xy).$$

Exercise 1.7. If $f(x,y) = \frac{2y}{y+\cos(x)}$, calculate $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.

$$\frac{\partial f}{\partial x} = \frac{2y\sin(x)}{(y+\cos(x))^2}, \quad \frac{\partial f}{\partial y} = \frac{2\cos(x)}{(y+\cos(x))^2}.$$

Exercise 1.8. Compute $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$, and $\frac{\partial f}{\partial z}$ if $f(x,y,z) = e^{xy}\ln z$.

$$\frac{\partial f}{\partial x} = ye^{xy}\ln z, \quad \frac{\partial f}{\partial y} = xe^{xy}\ln z, \quad \frac{\partial f}{\partial z} = \frac{e^{xy}}{z}.$$

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2 Implicit Partial Differentiation

Implicit partial differentiation in the multivariable setting works exactly as in the single variable case. Given a function $F(x,y,z)=0$ that defines z as a function of x and y, one computes $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ using the (single-variable) Chain Rule.

Example 2.1. If the equation

$$yz-\ln z = x+y$$

defines z implicitly as a function of x and y, then we compute $\frac{\partial z}{\partial x}$ by differentiating each side with respect to x:

$$\begin{aligned} y\frac{\partial z}{\partial x}-\frac{1}{z}\frac{\partial z}{\partial x} &= 1 \\ \Longrightarrow \frac{\partial z}{\partial x} &= \frac{1}{y-\frac{1}{z}}. \end{aligned}$$

Exercise 2.2. Compute $\frac{\partial z}{\partial y}$ if the equation

$$yz-\ln z = x+y$$

defines z implicitly as a function of x and y.

$$\frac{\partial z}{\partial y} = \frac{1-z}{y-\frac{1}{z}}.$$

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3 Higher-order partial derivatives

If f is a function of two variables, then each of its partial derivatives is also a function of two variables, so we can consider their partial derivatives. There are four possible second-order partial derivatives:

$$f_{xx} = \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right)$$
$$f_{xy} = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)$$
$$f_{yx} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right)$$
$$f_{yy} = \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right)$$

Exercise 3.1. Compute the second partial derivatives of

$$f(x,y) = x^3 + x^2y^3 - 2y^2.$$

SOLUTION In Example 1 we found that

$$f_x(x, y) = 3x^2 + 2xy^3 \qquad f_y(x, y) = 3x^2y^2 - 4y$$

Therefore

$$f_{xx} = \frac{\partial}{\partial x}(3x^2 + 2xy^3) = 6x + 2y^3 \qquad f_{xy} = \frac{\partial}{\partial y}(3x^2 + 2xy^3) = 6xy^2$$

$$f_{yx} = \frac{\partial}{\partial x}(3x^2y^2 - 4y) = 6xy^2 \qquad f_{yy} = \frac{\partial}{\partial y}(3x^2y^2 - 4y) = 6x^2y - 4$$

Notice for the function in Exercise 3.1 that $\frac{\partial f}{\partial x \partial y} = \frac{\partial f}{\partial y \partial x}$. This is no coincidence.

Theorem 3.2 (Equality of mixed partial derivatives)**.** *Let $f : U \subseteq \mathbb{R}^2 \to \mathbb{R}$. If $\frac{\partial f}{\partial x \partial y}$ and $\frac{\partial f}{\partial y \partial x}$ are both continuous on U, then $\frac{\partial f}{\partial x \partial y} = \frac{\partial f}{\partial y \partial x}$ throughout U.*

Proof. Let $h \neq 0$ and consider the quantity

$$\Delta(h) \equiv [f(a+h, b+h) - f(a+h, b)] - [f(a, b+h) - f(a, b)]. \tag{3.3}$$

If we let $g(x) \equiv f(x, b+h) - f(x, b)$, then

$$\Delta(h) = g(a+h) - g(a).$$

Since $\frac{\partial f}{\partial x}$ exists on U, g is differentiable on $(a, a+h)$. By the Mean Value Theorem, there exists $c \in (a, a+h)$ such that

$$\begin{aligned} g(a+h) - g(a) &= hg'(c) \\ &= h\left[\frac{\partial f}{\partial x}(c, b+h) - \frac{\partial f}{\partial x}(c, b)\right]. \end{aligned}$$

Since $\frac{\partial f}{\partial x}$ is differentiable with respect to y, applying the Mean Value Theorem again to $\frac{\partial f}{\partial x}$, there exists $d \in (b, b+h)$ such that

$$\frac{\partial f}{\partial x}(c, b+h) - \frac{\partial f}{\partial x}(c,b) = h\frac{\partial^2 f}{\partial y \partial x}(c,d).$$

Thus,

$$\Delta(h) = h^2 \frac{\partial^2 f}{\partial y \partial x}(c,d).$$

As $h \to 0$, $(c,d) \to (a,b)$, so by continuity of $\frac{\partial^2 f}{\partial y \partial x}$,

$$\lim_{h\to 0} \frac{\Delta(h)}{h^2} = \lim_{(c,d)\to(a,b)} \frac{\partial^2 f}{\partial y \partial x}(c,d) = \frac{\partial^2 f}{\partial y \partial x}(a,b).$$

Now regroup the terms in (3.3) as

$$\Delta(h) = [f(a+h, b+h) - f(a, b+h)] - [f(a+h, b) - f(a,b)]$$

and let $u(y) \equiv f(a+h, y) - f(a,y)$ so that $\Delta(h) = u(b+h) - u(b)$. By the Mean Value Theorem there exists $d \in (b, b+h)$ such that

$$\begin{aligned} u(b+h) - u(b) &= hg'(d) \\ &= h\left[\frac{\partial f}{\partial y}(a+h,d) - \frac{\partial f}{\partial y}(a,d)\right]. \end{aligned}$$

Applying the Mean Value Theorem again, there exists $c \in (a, a+h)$ such that

$$\frac{\partial f}{\partial y}(a+h,d) - \frac{\partial f}{\partial y}(a,d) = h\frac{\partial^2 f}{\partial x \partial y}(c,d),$$

so

$$\Delta(h) = h^2 \frac{\partial^2 f}{\partial x \partial y}(c,d).$$

Since $(c,d) \to (a,b)$ as $h \to 0$, by continuity of $\frac{\partial^2 f}{\partial x \partial y}$ we have

$$\lim_{h\to 0} \frac{\Delta(h)}{h^2} = \lim_{(c,d)\to(a,b)} \frac{\partial^2 f}{\partial x \partial y}(c,d) = \frac{\partial^2 f}{\partial x \partial y}(a,b).$$

Thus,

$$\frac{\partial^2 f}{\partial y \partial y}(a,b) = \frac{\partial^2 f}{\partial x \partial y}(a,b).$$

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Exercise 3.4. Compute the mixed partial derivatives of the function

$$f(x,y) = xy + e^y(y^2+1)$$

and verify that these are equal. In which order do we get the answer more quickly?

$f_x = y$, $f_{xy} = 1$, $f_y = x + e^y(y^2+1) + 2ye^y$, $f_{yx} = 1$. f_{xy} was faster, since the second term is independent of x.

Exercise 3.5. For each function below, determine which order of computing the mixed partial derivatives will give you the answer the fastest and compute them.

(a) $f(x,y) = x\sin y + e^y$

(b) $f(x,y) = 1/x$

(c) $f(x,y) = y + x^2y + 4y^3 - \ln(y^2+1)$

(d) $f(x,y) = x\ln(xy)$

(a) $f_x = \sin y$, $f_{xy} = \cos y$.

(b) $f_y = 0$, $f_{yx} = 0$.

(c) $f_x = 2xy$, $f_{xy} = 2x$.

(d) $f_y = \frac{x}{y}$, $f_{yx} = \frac{1}{y}$.

We can partially-differentiate a function as many times as we like as long as the derivatives exist:

$$\frac{\partial^3 f}{\partial y \partial y \partial x}(x,y), \frac{\partial^4 f}{\partial y \partial y \partial x \partial x}(x,y), \ldots.$$

By repeated application of Theorem 3.2, the order does not matter as long as the partial derivatives are all continuous.

Exercise 3.6. Compute

$$\frac{\partial^5 f}{\partial x \partial x \partial y \partial y \partial y}(x,y)$$

if $f(x,y) = y^5x^4e^x + 2\sin(3y)\cos(2y)$.

$f_x = y^5 4x^3e^x + y^5x^4e^x = y^5e^x(x^4+4x^3)$, $f_{xy} = 5y^4e^x(x^4+4x^3)$, $f_{xyy} = 20y^3e^x(x^4+4x^3)$, $f_{xyyy} = 60y^2e^x(x^4+4x^3)$, $f_{xyyyx} = 60y^2[e^x(x^4+4x^3) + e^x(4x^3+12x^2)] = 60y^2e^x(x^4+8x^3+12x^2)$.

The proof of Theorem 3.2 assumed continuity of the mixed partial derivatives. The next example shows that, if the mixed partial derivatives are not continuous, then Theorem 3.2 need not hold.

Example 3.7 (A function whose mixed partials are unequal)**.** Consider the following function

$$f(x,y) = \begin{cases} \frac{xy^3-x^3y}{x^2+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

For $(x,y) \neq (0,0)$, the quotient rule gives

$$\begin{aligned} \frac{\partial f}{\partial x} &= \frac{y^3 - 3x^2y}{x^2+y^2} - \frac{2x(xy^3-x^3y)}{(x^2+y^2)^2} \\ &= \frac{y^5 - 4x^2y^3 - x^4y}{(x^2+y^2)^2}. \end{aligned}$$

The partial derivative with respect to x at $(0,0)$ is

$$\frac{\partial f}{\partial x}(0,0) = \lim_{h\to 0} \frac{f(h,0)-f(0,0)}{h} = \frac{1}{h}\frac{0-0-0}{(h^2+0^2)^2} = 0,$$

so

$$\frac{\partial f}{\partial x} = \begin{cases} \frac{y^5-4x^2y^3-x^4y}{(x^2+y^2)^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0). \end{cases}$$

A similar calculation gives

$$\frac{\partial f}{\partial y} = \begin{cases} \frac{-x^5+4x^3y^2+xy^4}{(x^2+y^2)^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0). \end{cases}$$

The second partial derivative with respect to y is given by, when $(x,y) \neq (0,0)$

$$\frac{\partial f}{\partial y \partial x} = \frac{-x^6 - 9x^4y^2 + 9x^2y^4 + y^6}{(x^2+y^2)^3}$$

while at $(0,0)$ it is given by

$$\frac{\partial f}{\partial y \partial x}(0,0) = \lim_{h\to 0} \frac{\frac{\partial f}{\partial x}(0,h) - \frac{\partial f}{\partial x}(0,0)}{h} = \lim_{h\to 0} \frac{1}{h}\frac{h^5-0-0}{h^4} = \lim_{h\to 0} \frac{h}{h} = 1.$$

Thus,

$$\frac{\partial f}{\partial y \partial x} = \begin{cases} \frac{-x^6-9x^4y^2+9x^2y^4+y^6}{(x^2+y^2)^3}, & (x,y) \neq (0,0), \\ 1, & (x,y) = (0,0). \end{cases}$$

Now for $(x,y) \neq (0,0)$,

$$\frac{\partial f}{\partial x \partial y} = \frac{-x^6 - 9x^4y^2 + 9x^2y^4 + y^6}{(x^2+y^2)^3}$$

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and for $(x,y) = (0,0)$

$$\frac{\partial f}{\partial x \partial y}(0,0) = \lim_{h\to 0} \frac{1}{h} \frac{-h^5+0+0}{h^4} = \lim_{h\to 0} \frac{-h}{h} = -1.$$

Thus,

$$\frac{\partial f}{\partial x \partial y} = \begin{cases} \frac{-x^6-9x^4y^2+9x^2y^4+y^6}{(x^2+y^2)^3}, & (x,y) \neq (0,0) \\ -1, & (x,y) = (0,0). \end{cases}$$

We see that the mixed partial derivatives are clearly not both continuous at the origin (since they are equal for all $(x,y) \neq (0,0)$, to be continuous at the origin they would necessarily have the same value at the origin (which is the limit of this common function as $(x,y) \to (0,0)$), which they don't), and not equal there (though they are equal at all other points, as required by Theorem 3.2).

4 Partial Differential Equations

A *partial differential equation* (PDE) is an equation relating a function to one or more of its partial derivatives. PDEs arise frequently in physics and engineering. A famous one is *Laplace's equation*

$$\frac{\partial^2}{\partial x^2}u(x,y)+\frac{\partial^2}{\partial y^2}u(x,y)=0.$$

Solutions of Laplace's equation which are continuous are called *harmonic functions*. These play an important role in heat conduction, fluid flow, and electric potential.

Exercise 4.1. Show that the function $u(x,y)=e^x\sin y$ is harmonic.

$u_{xx}=u$, $u_{yy}=-u$, so $u_{xx}+u_{yy}=u-u=0$. Since u is also continuous, it is therefore harmonic.

Another important PDE in physics and engineering is the *wave equation*

$$\frac{\partial^2}{\partial t^2}u(x,t)-v^2\frac{\partial^2}{\partial x^2}u(x,t)=0,$$

which describes the motion of a wave propagating in the x-direction, such as a wave traveling on a vibrating string. The function $u(x,t)$ describes the displacement of the string at a distance x from one end of the string at time t. The constant v is the speed at which the wave propagates, which depends on the density of the string and the string tension.

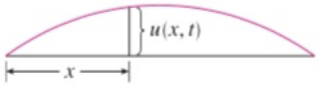


Figure 3: A solution $u(x,t)$ of the wave equation describes the displacement of a wave from equilibrium at position x and time t.

Exercise 4.2. Verify that the function $u(x,t)=A\sin(x-vt)$ is a solution of the wave equation for any real number A.

$u_{xx}=-u$, $u_t=vA\cos(x-vt)\implies u_{tt}=-v^2u$. Therefore $u_{tt}-v^2u_{xx}=-v^2u-v^2(-u)=0$. Since u is also continuous, it is therefore harmonic.